

MANDALGOVI, Mongolia

WMO: 443410

Lat: 45.7431N

Lon: 106.2642E

Elev: 1393

StdP: 85.67

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.1	-25.3	-32.4	0.2	-24.4	-30.3	0.3	-23.4	15.4	-18.9	13.6	-15.3	3.7	340	0.744

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	31.5	15.9	29.6	15.1	27.8	14.4	18.4	26.9	17.1	25.5	16.0	23.8	5.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	13.2	21.5	14.4	12.1	19.9	13.2	11.2	19.2	58.4	26.9	53.7	25.4	50.2	24.2	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
15.3	13.2	11.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-30.1	34.6	2.7	1.8	-32.0	35.9	-33.7	37.0	-35.2	38.0	-37.2	39.4
			-30.2	19.2	2.7	2.3	-32.1	20.9	-33.7	22.2	-35.2	23.5	-37.1	25.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	-16.8	-12.7	-4.1	4.8	11.7	18.1	21.0	18.8	12.5	3.3	-7.5	-14.8	(6)
(7)		DBStd	14.07	4.53	5.43	5.93	5.27	5.11	4.00	3.33	3.61	4.75	5.10	5.81	4.55	(7)
(8)		HDD10.0	3651	832	637	438	167	41	1	0	0	30	213	524	770	(8)
(9)		HDD18.3	5822	1090	870	696	405	211	52	12	38	181	466	774	1028	(9)
(10)		CDD10.0	1075	0	0	0	13	94	244	341	273	104	4	0	0	(10)
(11)		CDD18.3	204	0	0	0	0	6	46	94	53	5	0	0	0	(11)
(12)		CDH23.3	2071	0	0	0	2	113	508	916	470	62	0	0	0	(12)
(13)		CDH26.7	647	0	0	0	1	21	154	328	136	7	0	0	0	(13)
(14)	Wind	WSAvg	4.9	4.1	4.5	5.5	6.2	6.3	5.6	4.8	4.5	4.6	4.7	4.1	4.0	(14)
(15)	Precipitation	PrecAvg	135	1	1	2	4	10	24	38	41	11	5	2	1	(15)
(16)		PrecMax	248	6	10	7	19	47	93	103	114	57	44	8	6	(16)
(17)		PrecMin	58	0	0	0	0	0	1	0	0	0	0	0	0	(17)
(18)		PrecStd	48	1	2	2	5	10	18	22	27	12	8	2	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.1	3.4	14.7	22.3	29.1	33.0	35.0	32.7	27.8	19.5	9.4	-0.8	(19)
(20)		0.4%	MCWB	-5.7	-1.9	4.4	8.4	13.0	14.9	17.5	15.4	13.1	7.7	2.0	-4.6	(20)
(21)		2%	DB	-5.5	0.6	10.7	19.4	26.4	30.4	32.2	29.9	25.1	16.3	6.2	-3.8	(21)
(22)		2%	MCWB	-7.9	-3.8	2.2	7.4	11.3	14.1	17.0	15.4	12.1	6.6	0.3	-6.7	(22)
(23)		5%	DB	-7.7	-1.8	7.9	16.8	24.0	28.3	30.2	27.9	22.8	13.9	3.5	-6.0	(23)
(24)		5%	MCWB	-9.7	-5.3	0.9	6.2	10.1	13.9	16.0	14.4	11.3	5.2	-1.3	-8.2	(24)
(25)		10%	DB	-9.9	-4.1	5.2	14.4	21.6	26.2	28.2	25.9	20.7	11.5	1.0	-8.1	(25)
(26)		10%	MCWB	-11.5	-6.8	-0.7	4.8	9.2	13.1	15.4	14.0	10.2	3.9	-2.9	-9.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.4	-1.5	5.0	10.1	14.7	18.2	21.4	20.3	15.9	9.2	2.3	-4.4	(27)
(28)		0.4%	MCDWB	-2.8	2.5	13.2	20.0	25.9	26.3	29.8	28.5	24.3	17.1	8.8	-0.9	(28)
(29)		2%	WB	-7.9	-3.6	2.7	8.0	12.5	16.5	19.0	17.5	13.7	7.1	0.5	-6.5	(29)
(30)		2%	MCDWB	-5.7	0.3	9.9	17.5	22.7	25.5	27.9	25.2	22.0	15.5	6.1	-4.0	(30)
(31)		5%	WB	-9.7	-5.2	1.1	6.6	10.9	15.3	17.7	16.3	12.2	5.6	-1.2	-8.1	(31)
(32)		5%	MCDWB	-7.7	-2.0	7.5	16.4	22.2	24.4	26.4	23.6	20.5	13.1	3.2	-6.0	(32)
(33)		10%	WB	-11.5	-6.8	-0.6	5.1	9.6	14.2	16.6	15.3	11.0	4.2	-2.7	-9.9	(33)
(34)		10%	MCDWB	-9.9	-4.2	5.2	13.7	20.5	23.5	25.0	22.0	19.5	11.1	0.9	-8.1	(34)
(35)	Mean Daily Temperature Range		MDBR	9.9	11.7	12.5	12.7	12.9	11.8	11.0	10.8	11.8	11.8	10.7	9.5	(35)
(36)		5% DB	MCDBR	12.1	14.3	15.2	16.0	15.9	14.4	13.7	13.4	14.2	14.3	12.9	11.4	(36)
(37)		5% DB	MCWBR	10.1	10.9	9.5	8.8	7.7	5.9	4.9	5.2	6.6	8.2	8.8	9.1	(37)
(38)		5% WB	MCDBR	12.0	13.8	14.5	15.1	14.3	12.1	11.4	10.7	12.5	13.7	12.4	11.1	(38)
(39)		5% WB	MCWBR	10.1	10.7	9.5	8.6	7.7	5.8	5.3	5.3	6.4	8.2	8.6	8.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.243	0.268	0.305	0.351	0.377	0.397	0.377	0.359	0.321	0.287	0.256	0.241	(40)	
(41)		taud	2.379	2.326	2.268	2.187	2.156	2.193	2.311	2.337	2.391	2.419	2.411	2.381	(41)	
(42)		Ebn at Noon	885	920	924	905	888	869	883	887	898	886	859	847	(42)	
(43)		Edh at Noon	71	92	114	136	146	142	124	116	100	83	68	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.05	3.15	4.77	6.02	7.02	6.91	6.58	5.99	5.03	3.62	2.32	1.70	(44)	
(45)		RadStd	0.17	0.23	0.16	0.30	0.34	0.35	0.27	0.25	0.17	0.14	0.15	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-2.46	N/A	-1.12	-1.19	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	-1.87	N/A	-1.01	-0.85	N/A	N/A	N/A	N/A

Nomenclature: See separate page